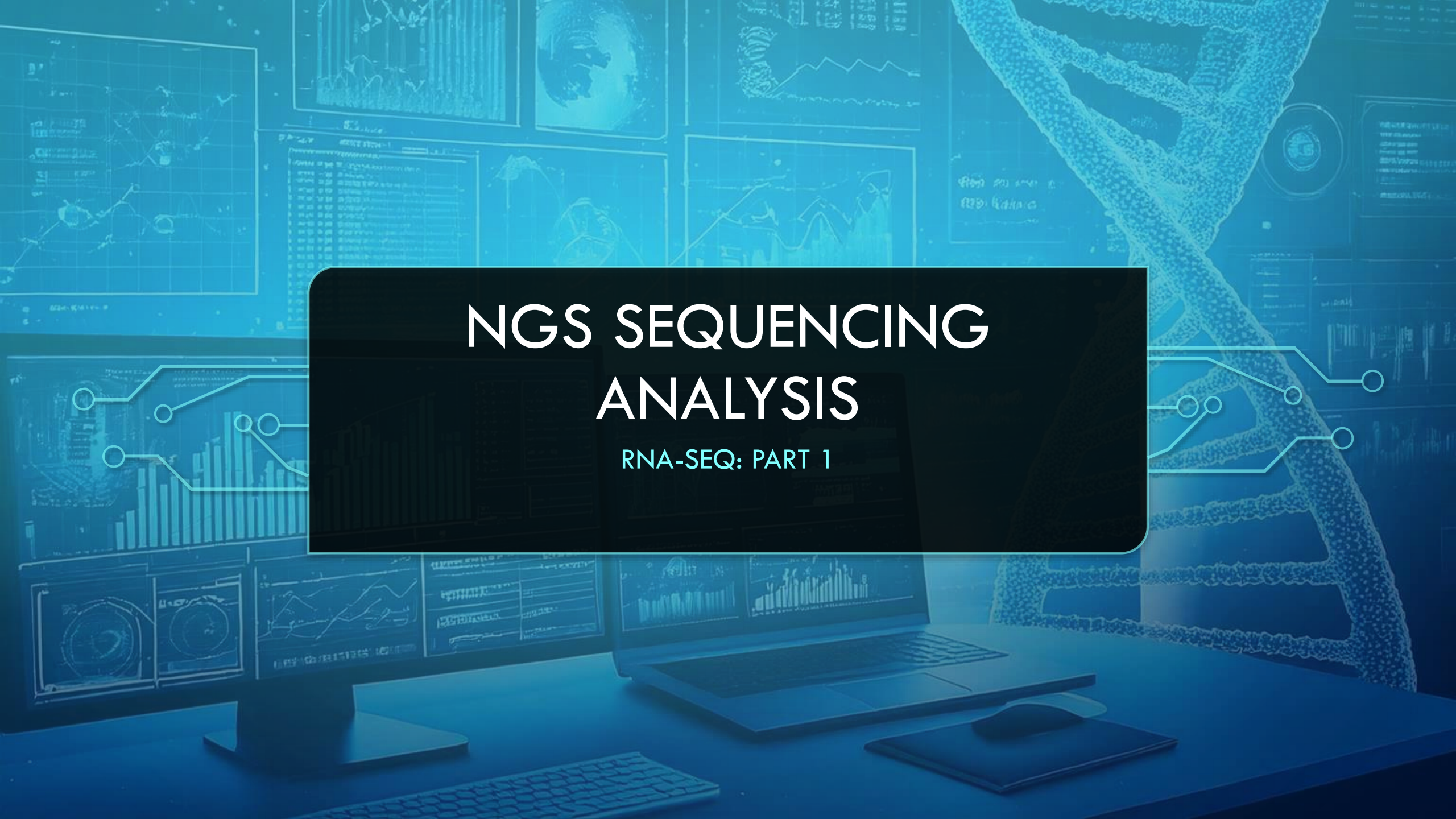




NGS SEQUENCING ANALYSIS

RNA-SEQ: PART 1

REVIEW

- Last time working in Linux, we downloaded RNA-seq files from the Gene Expression Omnibus (GEO)
- We used a combination of the tools “prefetch” and “fasterq-dump” from the SRA Toolkit
- We automated this process using a “for loop” and submitted a “batch job” to run on the cluster

The screenshot shows the NCBI GEO website interface. At the top, there are logos for NCBI and GEO (Gene Expression Omnibus). Navigation links include HOME, SEARCH, SITE MAP, GEO Publications, FAQ, MIAME, and Email GEO. The breadcrumb trail indicates the current location: NCBI > GEO > Accession Display. A search bar at the top right shows the accession number GSE270286. Below the search bar, there are dropdown menus for Scope (Self), Format (HTML), and Amount (Quick), followed by a GO button. The main content area displays the details for Series GSE270286. The status is 'Public on Nov 06, 2024'. The title is 'Transcriptomic analysis of cellular senescence induced by ectopic expression of ATF6a in human breast cancer cells'. The organism is 'Homo sapiens'. The experiment type is 'Expression profiling by high throughput sequencing'. The summary describes the study: 'To investigate the molecular mechanism that triggers endoplasmic reticulum (ER) stress-induced cellular senescence in breast cancer cells, we established MCF7 cell lines exposed to ATF6a ectopic expression. We then performed gene expression profiling analysis using data obtained from RNA-seq of 9 different cells at three time points.' The overall design is 'Comparative gene expression profiling analysis of RNA-seq data for MCF-7 cells from three distinct time points (baseline, 6 hours, and 9 hours after ATF6a ectopic expression), using Illumina HiSeq X10 and NovaSeq 6000 platforms with paired-end 100 base sequencing.' The contributor(s) are 'Kim J, Bae S, Moon Y, Kim E, Kim Y, Park Y, Han M, Sohn J'. The citation(s) are 'Kim JW, Bae SH, Moon Y, Kim EK et al. Transcriptomic analysis of cellular senescence induced by ectopic expression of ATF6a in human breast cancer cells. PLoS One 2024;19(10):e0309749. PMID: 39466820'.

Status	Public on Nov 06, 2024
Title	Transcriptomic analysis of cellular senescence induced by ectopic expression of ATF6a in human breast cancer cells
Organism	Homo sapiens
Experiment type	Expression profiling by high throughput sequencing
Summary	To investigate the molecular mechanism that triggers endoplasmic reticulum (ER) stress-induced cellular senescence in breast cancer cells, we established MCF7 cell lines exposed to ATF6a ectopic expression. We then performed gene expression profiling analysis using data obtained from RNA-seq of 9 different cells at three time points.
Overall design	Comparative gene expression profiling analysis of RNA-seq data for MCF-7 cells from three distinct time points (baseline, 6 hours, and 9 hours after ATF6a ectopic expression), using Illumina HiSeq X10 and NovaSeq 6000 platforms with paired-end 100 base sequencing.
Contributor(s)	Kim J , Bae S , Moon Y , Kim E , Kim Y , Park Y , Han M , Sohn J
Citation(s)	Kim JW, Bae SH, Moon Y, Kim EK et al. Transcriptomic analysis of cellular senescence induced by ectopic expression of ATF6a in human breast cancer cells. <i>PLoS One</i> 2024;19(10):e0309749. PMID: 39466820

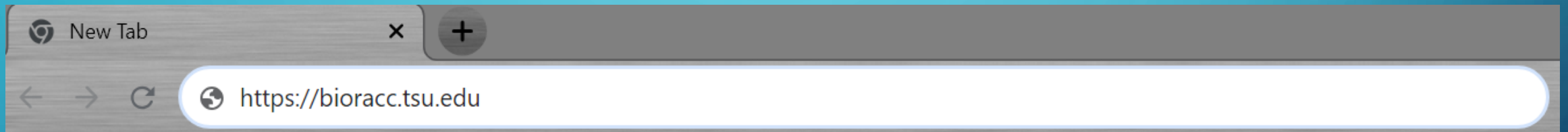
LOG ON TO THE CLUSTER

OPEN

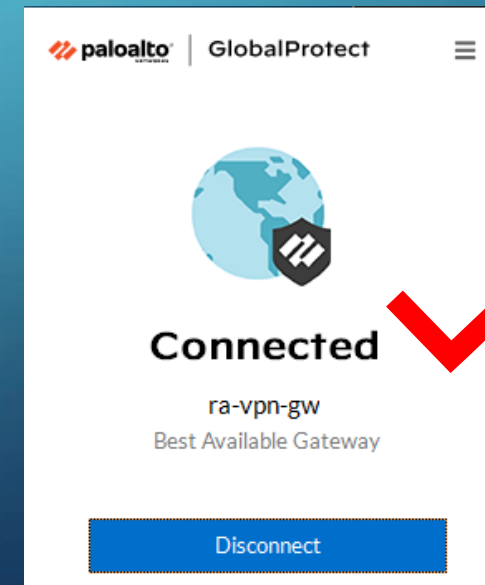
OnDemand

OnDemand provides an integrated, single access point for all of your HPC resources.

1. Type <https://bioracc.tsu.edu> in a browser



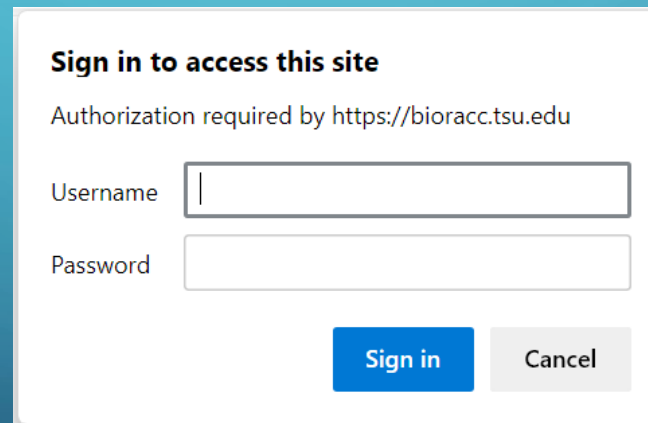
Must be connected to TSU-NET secure network or the VPN to access cluster



CREDENTIALS

Username: firstname.lastname (all lowercase)

Default password: firstname.lastname.**changeme**



Sign in to access this site

Authorization required by <https://bioracc.tsu.edu>

Username

Password

*Did you change your password last time?

SUCCESSFULLY LOGGED IN



TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

 ▾  



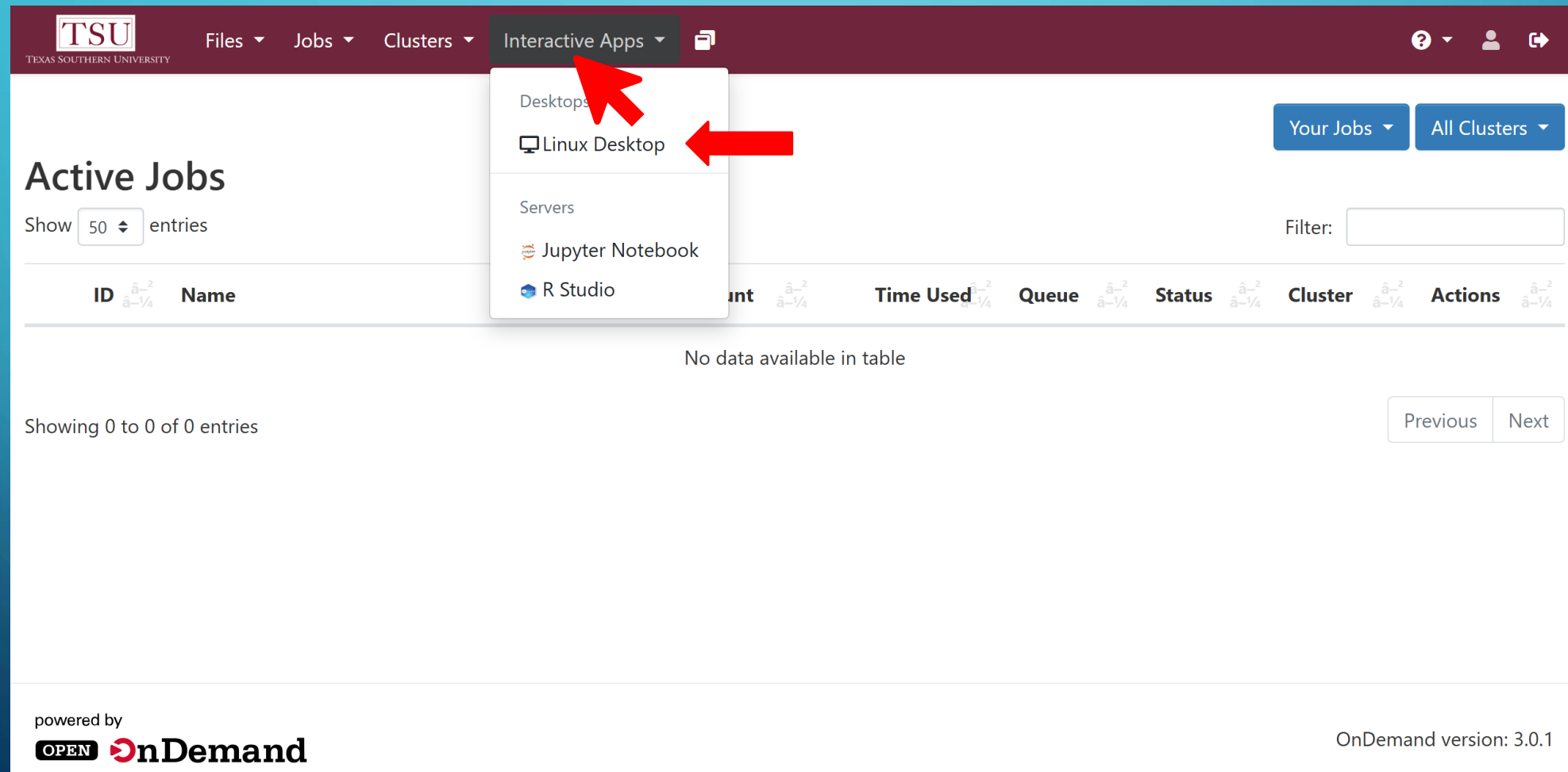
Solutions through the Lens of Science

OnDemand provides an integrated, single access point for all of your HPC resources.

powered by
 

OnDemand version: 3.0.1

REQUESTING A LINUX DESKTOP SESSION



The screenshot displays the TSU OnDemand web interface. The top navigation bar includes links for Files, Jobs, Clusters, and Interactive Apps. The 'Interactive Apps' dropdown menu is open, showing options for Desktops, Linux Desktop, Servers, Jupyter Notebook, and R Studio. Two red arrows point to the 'Linux Desktop' option. The main content area shows 'Active Jobs' with a table header and a message indicating no data is available. The footer includes the TSU logo, the OnDemand logo, and the version number 3.0.1.

TSU
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

Desktops
Linux Desktop
Servers
Jupyter Notebook
R Studio

Your Jobs ▾ All Clusters ▾

Active Jobs

Show 50 entries

Filter:

ID	Name	Amount	Time Used	Queue	Status	Cluster	Actions
No data available in table							

Showing 0 to 0 of 0 entries

Previous Next

powered by
OPEN OnDemand

OnDemand version: 3.0.1

REQUESTING A LINUX DESKTOP SESSION

 **TSU**
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

? ▾  

Home / My Interactive Sessions / Linux Desktop

Interactive Apps

Desktops

 Linux Desktop

Servers

 Jupyter Notebook

 R Studio

Linux Desktop

This app will launch an interactive desktop on a compute node. You will have full access to the resources defined below. This is analogous to an interactive batch job.

How many resources do you need for this desktop?

☒ 4 Cores 

☐ 8 Cores

☐ 16 Cores, 1 GPU

☐ 32 Cores, 1 GPU

How many hours do you need the resources for?



 **Launch** 

* The Linux Desktop session data for this session can be accessed under the [data root directory](#).

REQUESTING A LINUX DESKTOP SESSION

TSU
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

? ▾  

Session was successfully created. 

Home / My Interactive Sessions

Interactive Apps

Desktops

 Linux Desktop

Servers

 Jupyter Notebook

 R Studio

Linux Desktop (4201)

Queued

Created at: 2025-01-26 12:33:43 CST

Time Requested: 2 hours

Session ID: [cf54a055-29d2-411d-9a83-e3fa32633f77](#)



Please be patient as your job currently sits in queue. The wait time depends on the number of cores as well as time requested.



powered by
 

OnDemand version: 3.0.1

LAUNCH LINUX DESKTOP SESSION



TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

 ▾  

Session was successfully created. 

Home / My Interactive Sessions

Interactive Apps

Desktops

 Linux Desktop

Servers

 Jupyter Notebook

 R Studio

Linux Desktop (4201) 1 node | 4 cores | Running 

Host: >_gpu001.cm.cluster  Delete

Created at: 2025-01-26 12:33:43 CST

Time Remaining: 1 hour and 59 minutes

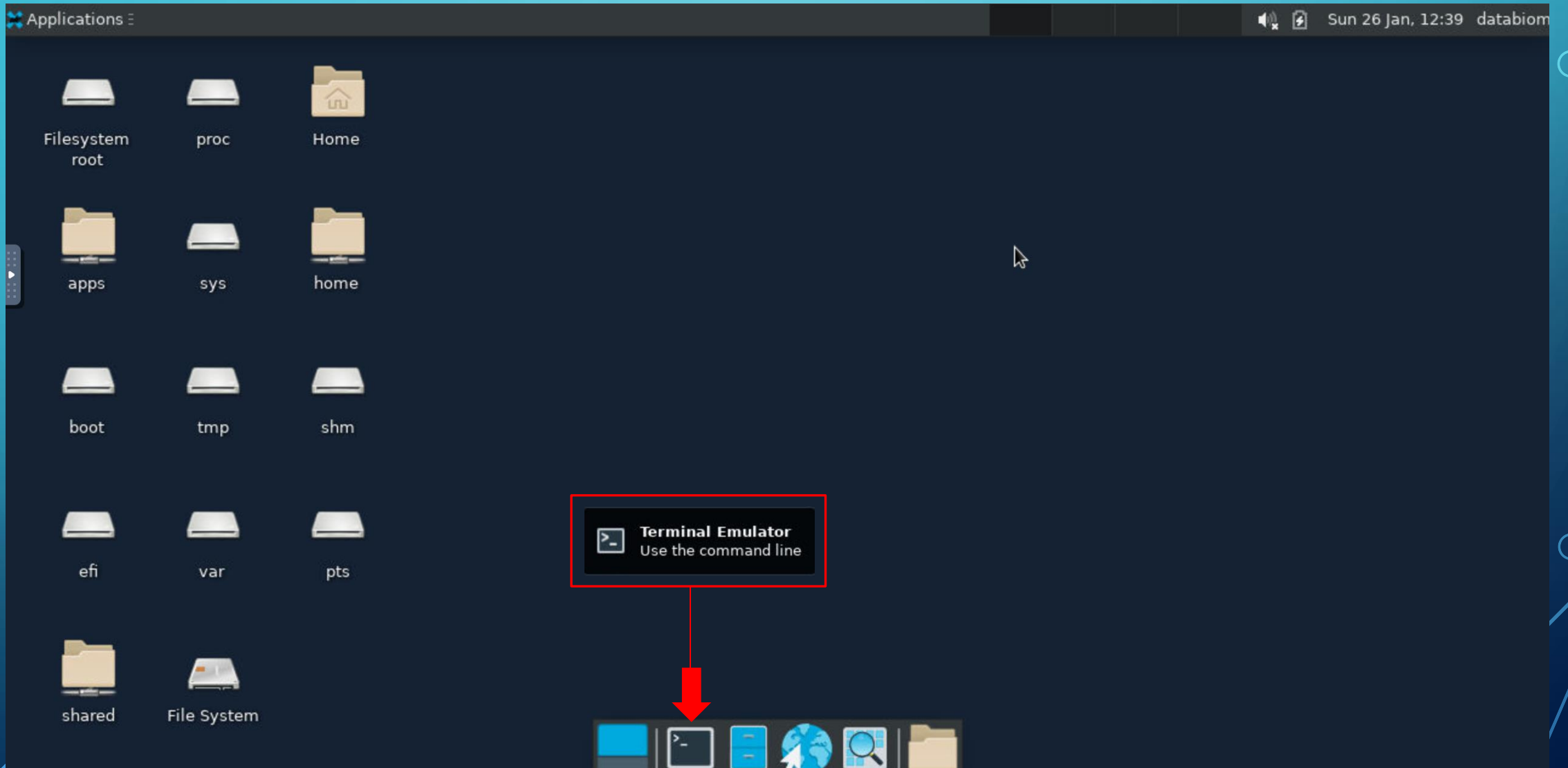
Session ID: cf54a055-29d2-411d-9a83-e3fa32633f77

Compression  0 (low) to 9 (high)

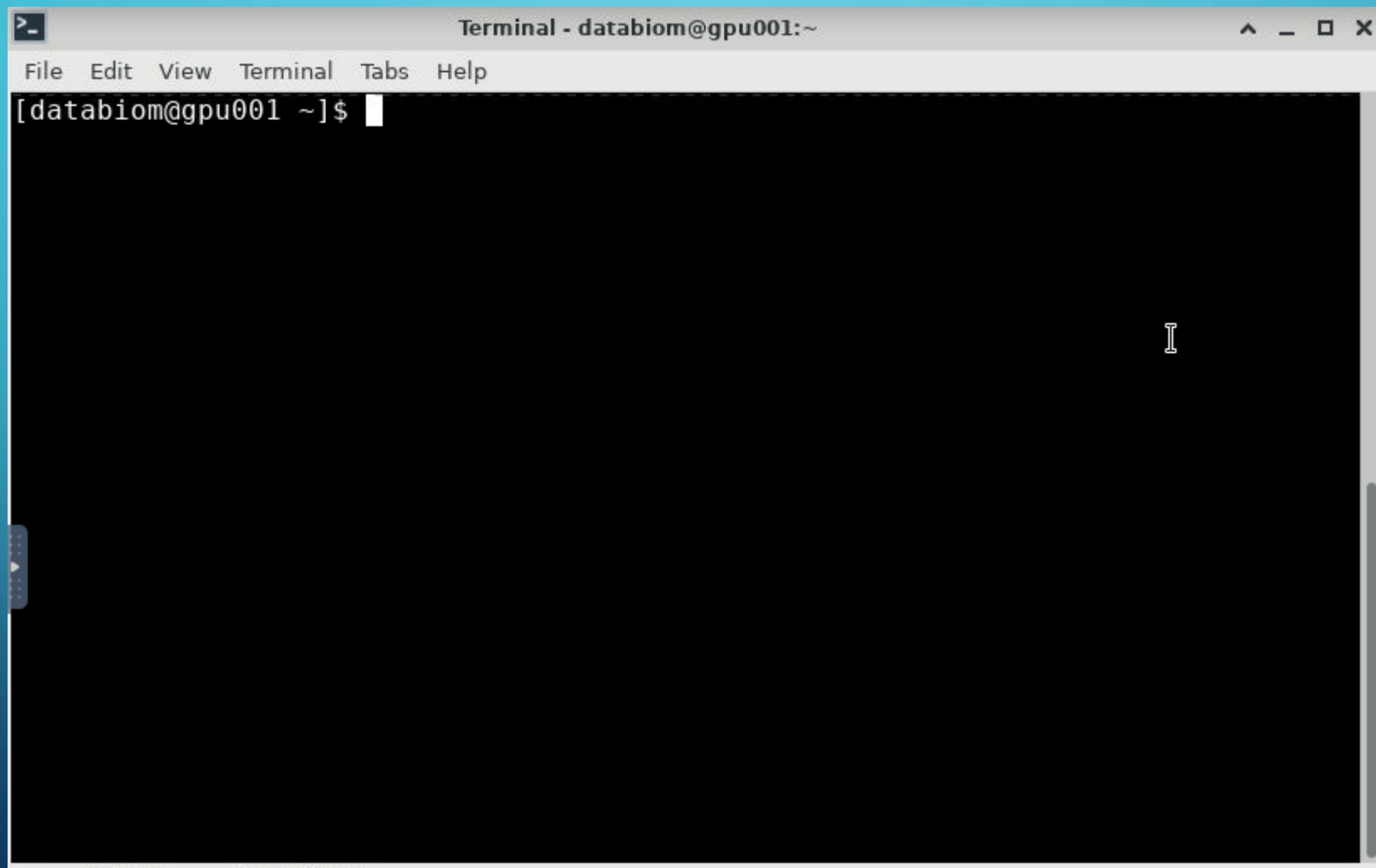
Image Quality  0 (low) to 9 (high)

 Launch Linux Desktop View Only (Share-able Link)

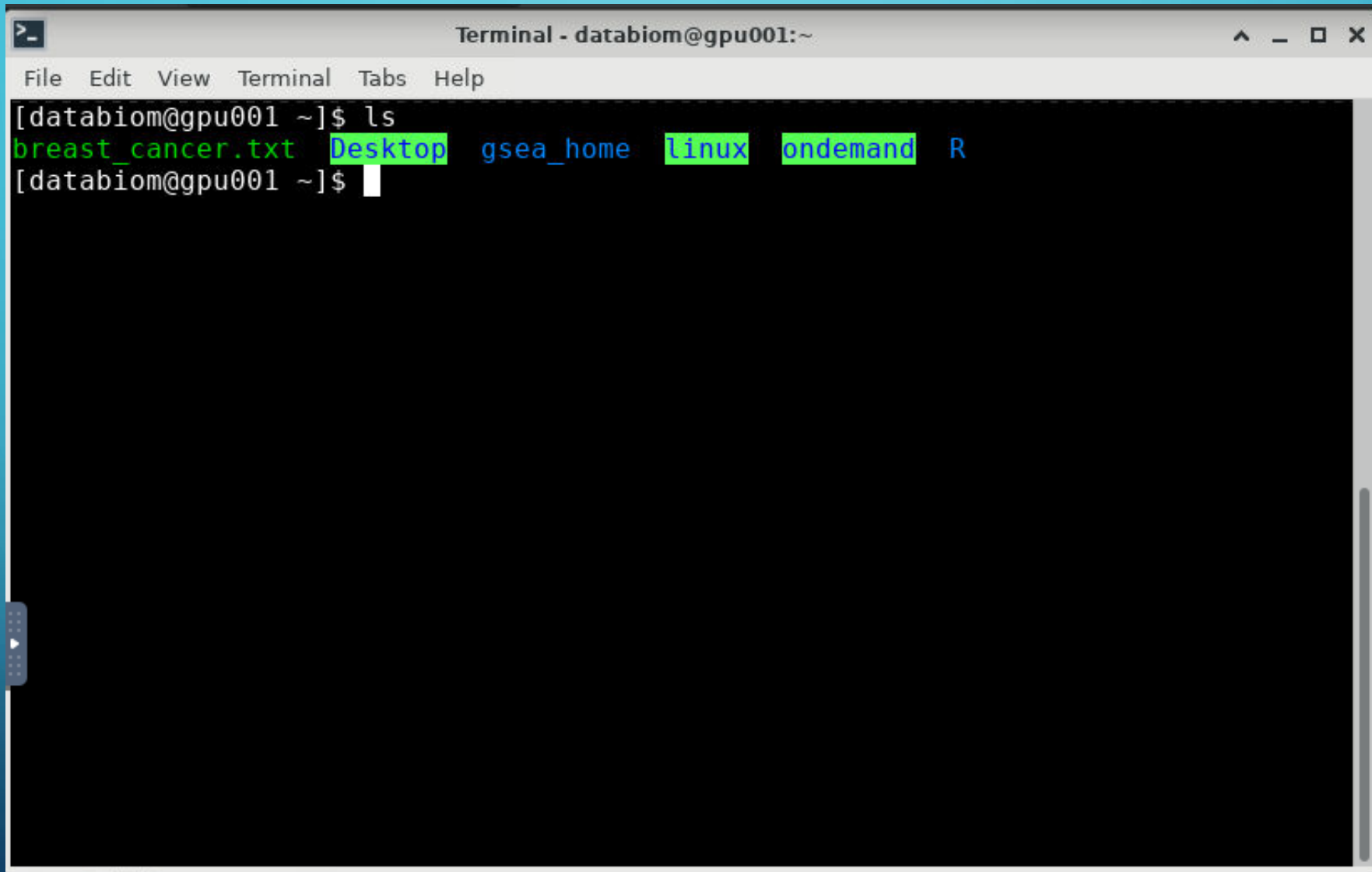
OPEN A NEW TERMINAL



TERMINAL



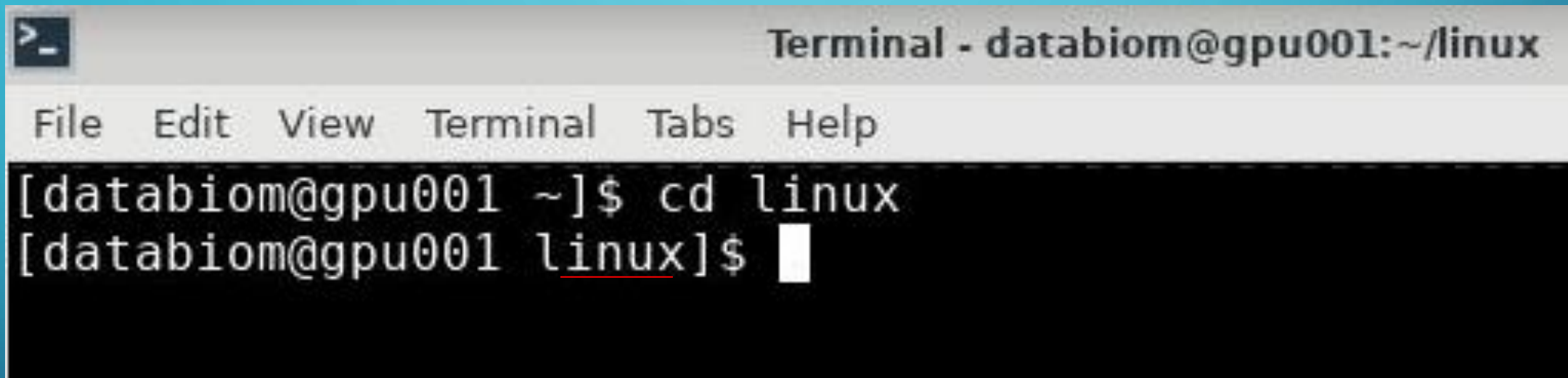
LIST CONTENTS OF HOME DIRECTORY



```
Terminal - databiom@gpu001:~  
File Edit View Terminal Tabs Help  
[databiom@gpu001 ~]$ ls  
breast_cancer.txt Desktop gsea_home linux ondemand R  
[databiom@gpu001 ~]$
```

The image shows a terminal window titled "Terminal - databiom@gpu001:~". The window has a menu bar with "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal content shows the user running the command `ls` in their home directory. The output lists several files and directories: `breast_cancer.txt`, `Desktop`, `gsea_home`, `linux`, `ondemand`, and `R`. The prompt `[databiom@gpu001 ~]$` is shown again on the next line, indicating the command has been executed.

CHANGE (GO INTO) THE 'LINUX' DIRECTORY

A terminal window titled "Terminal - databiom@gpu001:~/linux". The window has a menu bar with "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal content shows the command "cd linux" being executed, changing the directory from "~" to "linux". The prompt changes from "[databiom@gpu001 ~]" to "[databiom@gpu001 linux]".

```
>_ Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 ~]$ cd linux
[databiom@gpu001 linux]$
```

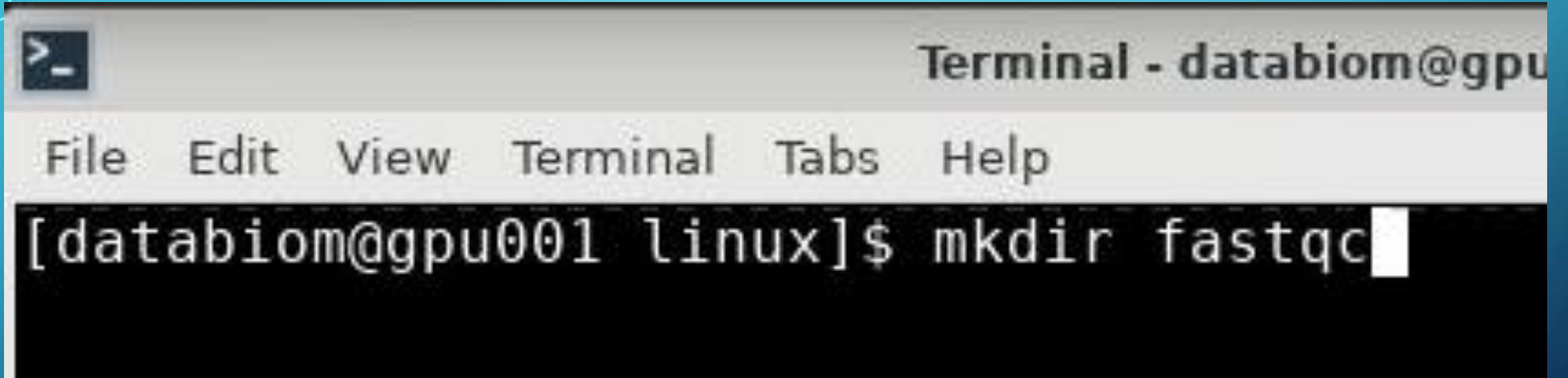
*Did you create a 'linux' directory last time?

CONTENTS OF 'LINUX' DIRECTORY

```
Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 ~]$ cd linux/
[databiom@gpu001 linux]$
[databiom@gpu001 linux]$ ls
breast_cancer.txt  GSE270286.txt          SRR29476514  SRR29476521
fastqc             main_job.sh            SRR29476515  SRR29476522
fastqc.sh          mutation_types.txt     SRR29476516  Y537S.txt
GSE270286_sra.out  sratoolkit_prefetch_dump_batch.sh SRR29476520
[databiom@gpu001 linux]$
```

These are the sequencing files we
downloaded previously

MAKE A NEW DIRECTORY FOR OUTPUT

A terminal window titled "Terminal - databiom@gpu" with a menu bar containing "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal shows the command "[databiom@gpu001 linux]\$ mkdir fastqc" being entered, with a white cursor at the end of the command.

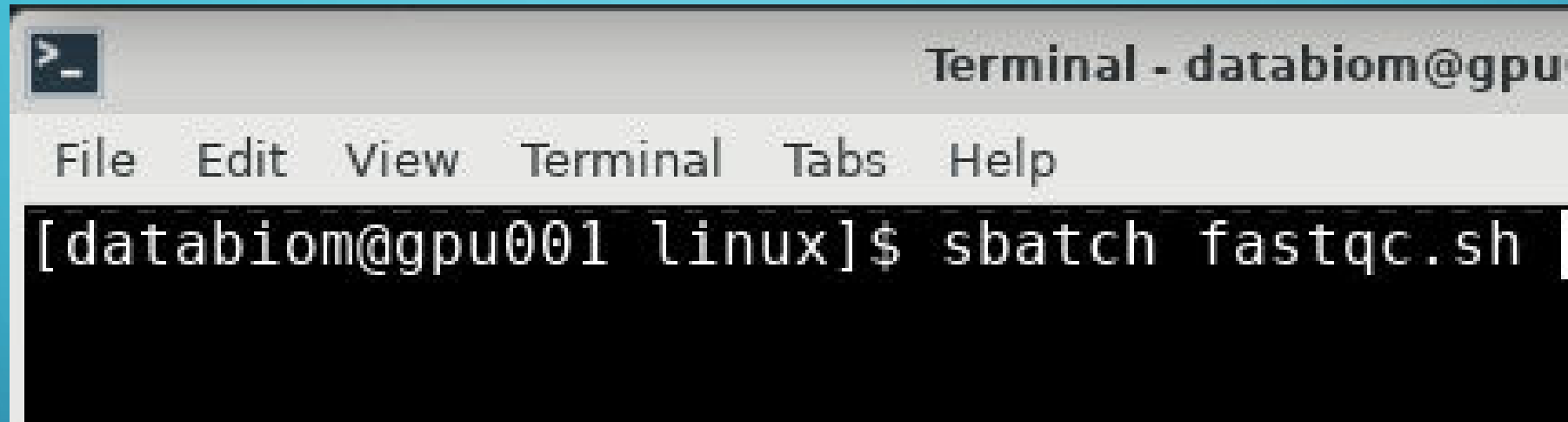
```
>_ Terminal - databiom@gpu
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ mkdir fastqc
```

COPY OVER THE FASTQC.SH SCRIPT TO RUN QC

```
Terminal - databiom@gpu001:~/linux  
File Edit View Terminal Tabs Help  
[databiom@gpu001 linux]$ cp /home/databiom/linux/fastqc.sh .
```

Don't forget

SUBMIT SCRIPT USING 'SBATCH'



```
Terminal - databiom@gpu001
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ sbatch fastqc.sh
```

CHECK TO MAKE SURE YOUR JOB IS RUNNING USING 'SQUEUE'

```
Terminal - databiom@gpu001: ~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ squeue
SON)
      JOBID PARTITION      NAME      USER ST       TIME  NODES NODELIST
      4745      defq  ::app=Li aleye.ig    R    6:10:28      1 gpu001
      4747      defq  ::app=Li databiom  R    4:50:49      1 gpu001
      4748      defq  ::app=Li grace.ol    R    4:31:07      1 gpu001
      4749      defq  ::app=Li      ceci    R    4:09:44      1 gpu001
      4759      defq  ::app=Li aleye.ig    R    1:22:53      1 gpu001
      4761      defq  ::app=Li roshitha    R    1:00:02      1 gpu001
      4762      defq  ::app=Li aleye.ig    R     55:16      1 gpu001
      4763      defq  ::app=Li aleye.ig    R     50:22      1 gpu001
      4767      defq  ::app=Li krishna.    R     28:58      1 gpu003
      4772      defq  ::app=Li aleye.ig    R      3:05      1 gpu003
```



Should take ~12 minutes



NEXT-GENERATION SEQUENCING (NGS)

- Next-generation sequencing (NGS) is a massively parallel sequencing technology that offers ultra-high throughput, scalability, and speed.
- The technology is used to determine the order of nucleotides in entire genomes or targeted regions of DNA or RNA.
- NGS has revolutionized the biological sciences, allowing labs to perform a wide variety of applications and study biological systems at a level never before possible.

APPLICATIONS OF NGS

- Rapidly sequence whole genomes
- Deeply sequence target regions
- Utilize RNA sequencing (RNA-Seq) to discover novel RNA variants and splice sites, or quantify mRNAs for gene expression analysis
- Analyze epigenetic factors such as genome-wide DNA methylation and DNA-protein interactions
- Sequence cancer samples to study rare somatic variants, tumor subclones, and more
- Study the human microbiome
- Identify novel pathogens

WORKFLOW

NGS workflow

The next-generation sequencing workflow includes three basic steps: library preparation, sequencing, and data analysis.



Library preparation



Sequencing



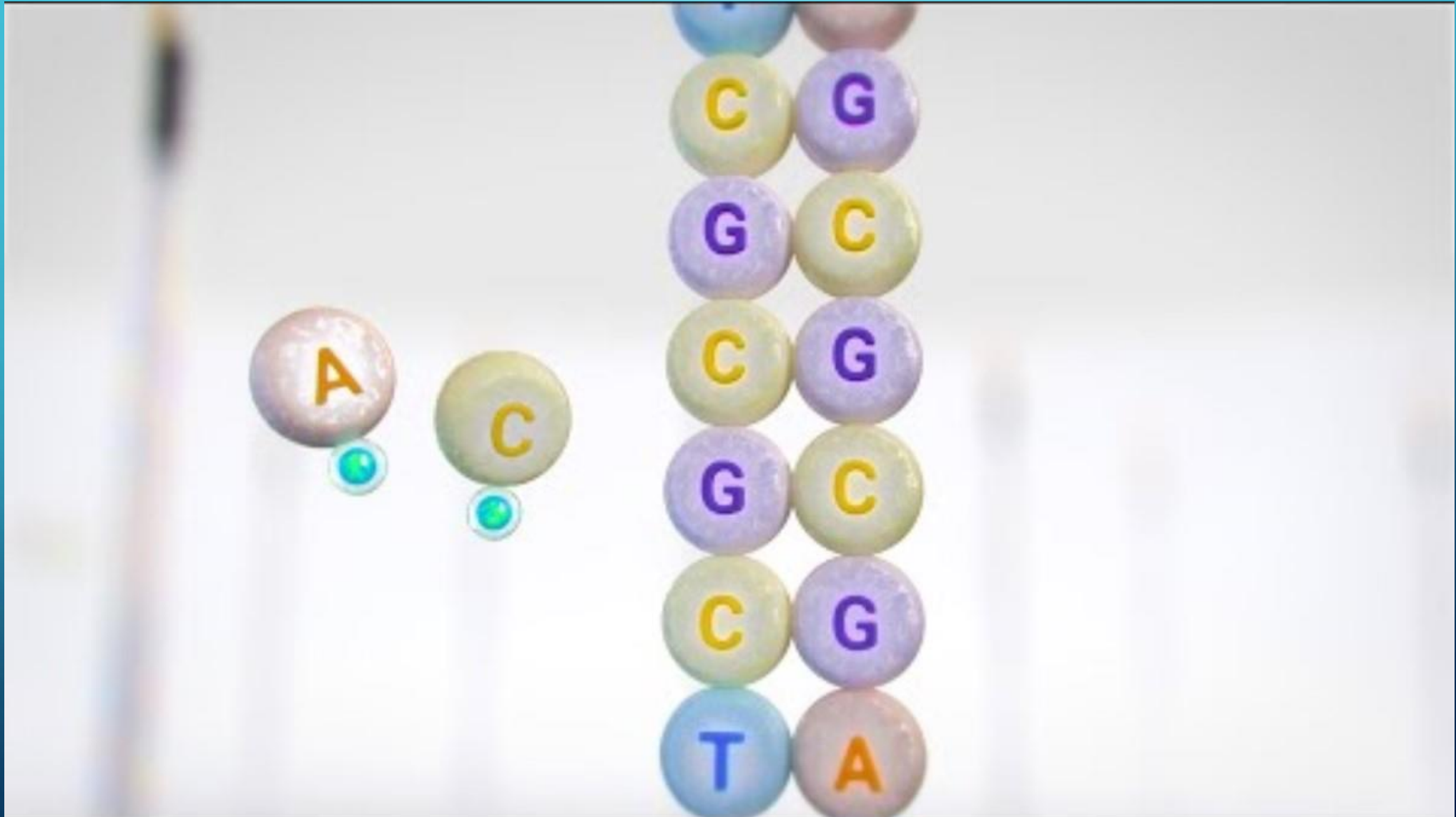
Data analysis

NGS library prep education

NGS platforms

NGS data analysis tools

HOW SEQUENCING IS PERFORMED



QUALITY CONTROL (QC) OF SEQUENCING READS



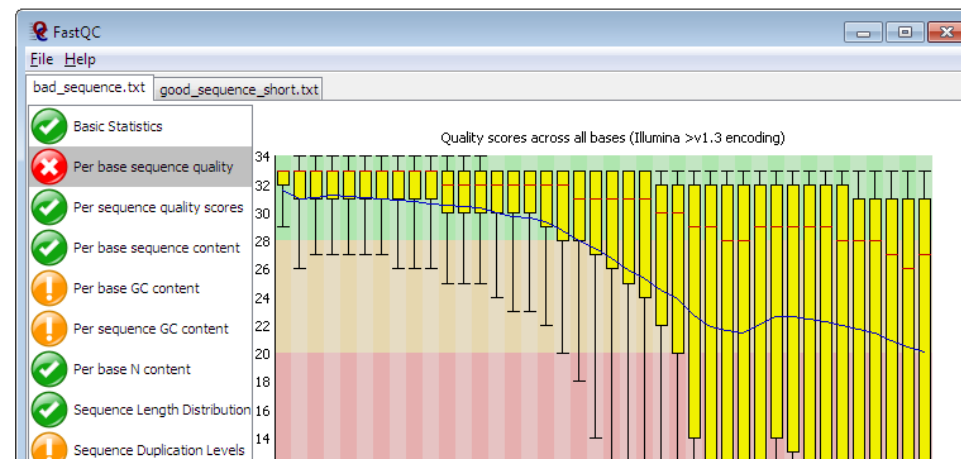
Babraham Bioinformatics

[About](#) | [People](#) | [Services](#) | [Projects](#) | [Training](#) | [Publications](#)

FastQC

Function	A quality control tool for high throughput sequence data.
Language	Java
Requirements	A suitable Java Runtime Environment The Picard BAM/SAM Libraries (included in download)
Code Maturity	Stable. Mature code, but feedback is appreciated.
Code Released	Yes, under GPL v3 or later .
Initial Contact	Simon Andrews

[Download Now](#)



QUALITY CONTROL (QC) METRICS

- Per Base Sequence
- Per Sequence Quality
- Per Base Sequence Content
- Per Sequence GC Content
- Per Base N Content
- Sequence Length Distribution
- Duplicate Sequences
- Sequence Length Distribution
- Duplicate Sequences
- Overrepresented Sequences
- Adapter Content
- Kmer Content
- Per Tile Sequence Quality

FASTQC OUTPUT

After the fastqc job has finished, you should have the results in the 'fastqc' directory that you created. 'cd' into the 'fastqc' directory and view the contents:

```
Terminal - databiom@gpu001:~/linux/fastqc
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ cd fastqc/
[databiom@gpu001 fastqc]$
[databiom@gpu001 fastqc]$ ls
SRR29476514_1_fastqc.html  SRR29476516_1_fastqc.html  SRR29476521_1_fastqc.html
SRR29476514_1_fastqc.zip  SRR29476516_1_fastqc.zip  SRR29476521_1_fastqc.zip
SRR29476514_2_fastqc.html  SRR29476516_2_fastqc.html  SRR29476521_2_fastqc.html
SRR29476514_2_fastqc.zip  SRR29476516_2_fastqc.zip  SRR29476521_2_fastqc.zip
SRR29476515_1_fastqc.html  SRR29476520_1_fastqc.html  SRR29476522_1_fastqc.html
SRR29476515_1_fastqc.zip  SRR29476520_1_fastqc.zip  SRR29476522_1_fastqc.zip
SRR29476515_2_fastqc.html  SRR29476520_2_fastqc.html  SRR29476522_2_fastqc.html
SRR29476515_2_fastqc.zip  SRR29476520_2_fastqc.zip  SRR29476522_2_fastqc.zip
[databiom@gpu001 fastqc]$
```

MULTIQC TOOL

We can use another tool called “multiqc” to combine all the results for the individual runs into one report for convenience



Aggregate bioinformatics results across many samples into a single report

MULTIQC OVERVIEW

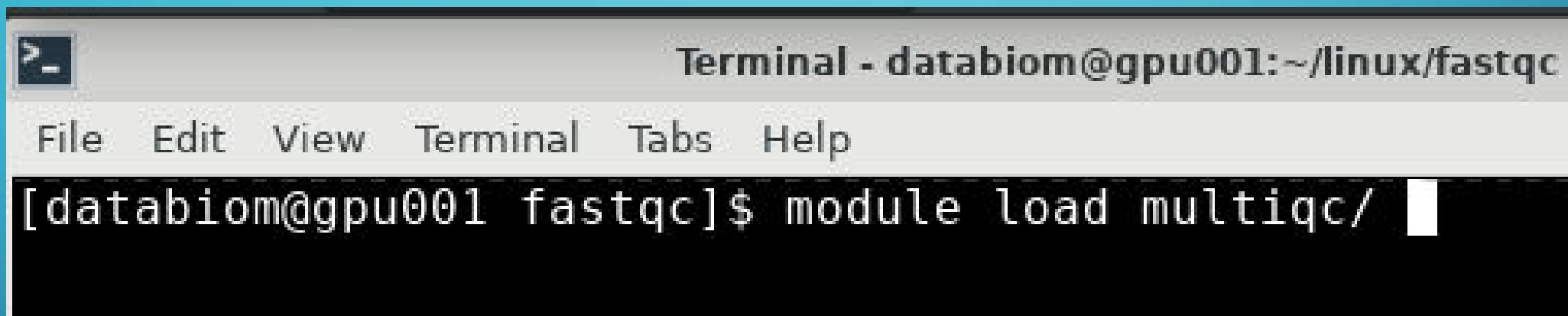


Using MultiQC Reports

Phil Ewels
phil.ewels@scilifelab.se

LOAD MULTIQC MODULE

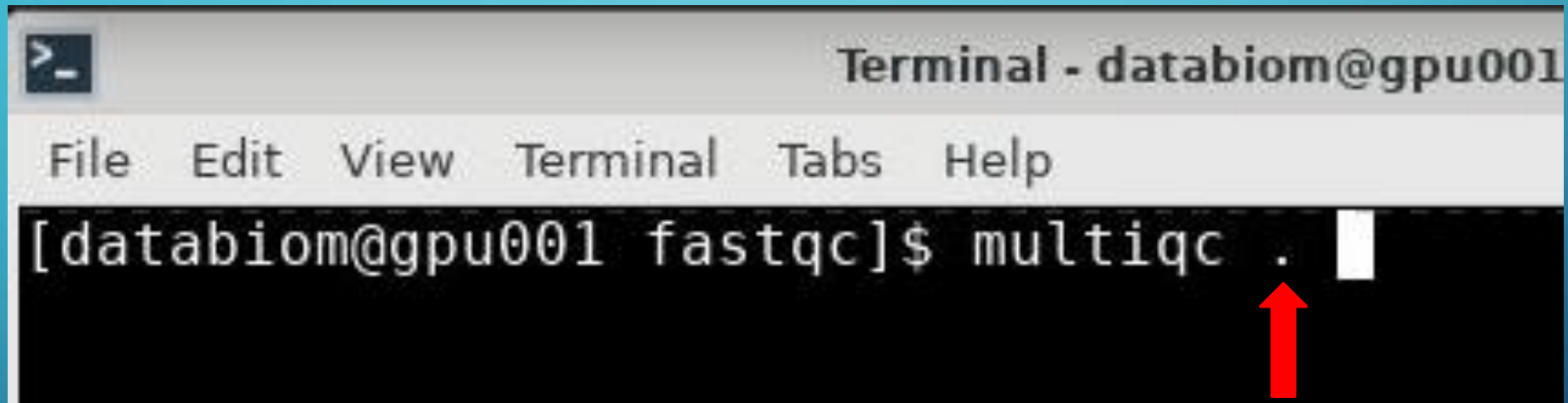
First, load the 'multiqc' module:



```
Terminal - databiom@gpu001:~/linux/fastqc
File Edit View Terminal Tabs Help
[databiom@gpu001 fastqc]$ module load multiqc/
```

RUNNING MULTIQC

Then, run multiqc within the 'fastqc' directory:



```
Terminal - databiom@gpu001
File Edit View Terminal Tabs Help
[databiom@gpu001 fastqc]$ multiqc .
```

A terminal window titled "Terminal - databiom@gpu001" with a menu bar containing "File", "Edit", "View", "Terminal", "Tabs", and "Help". The command prompt shows the user is in the "fastqc" directory and has entered the command "multiqc .". A red arrow points to the period at the end of the command.



You must be in the 'fastqc' directory

MULTIQC

You should see this if it successfully ran:

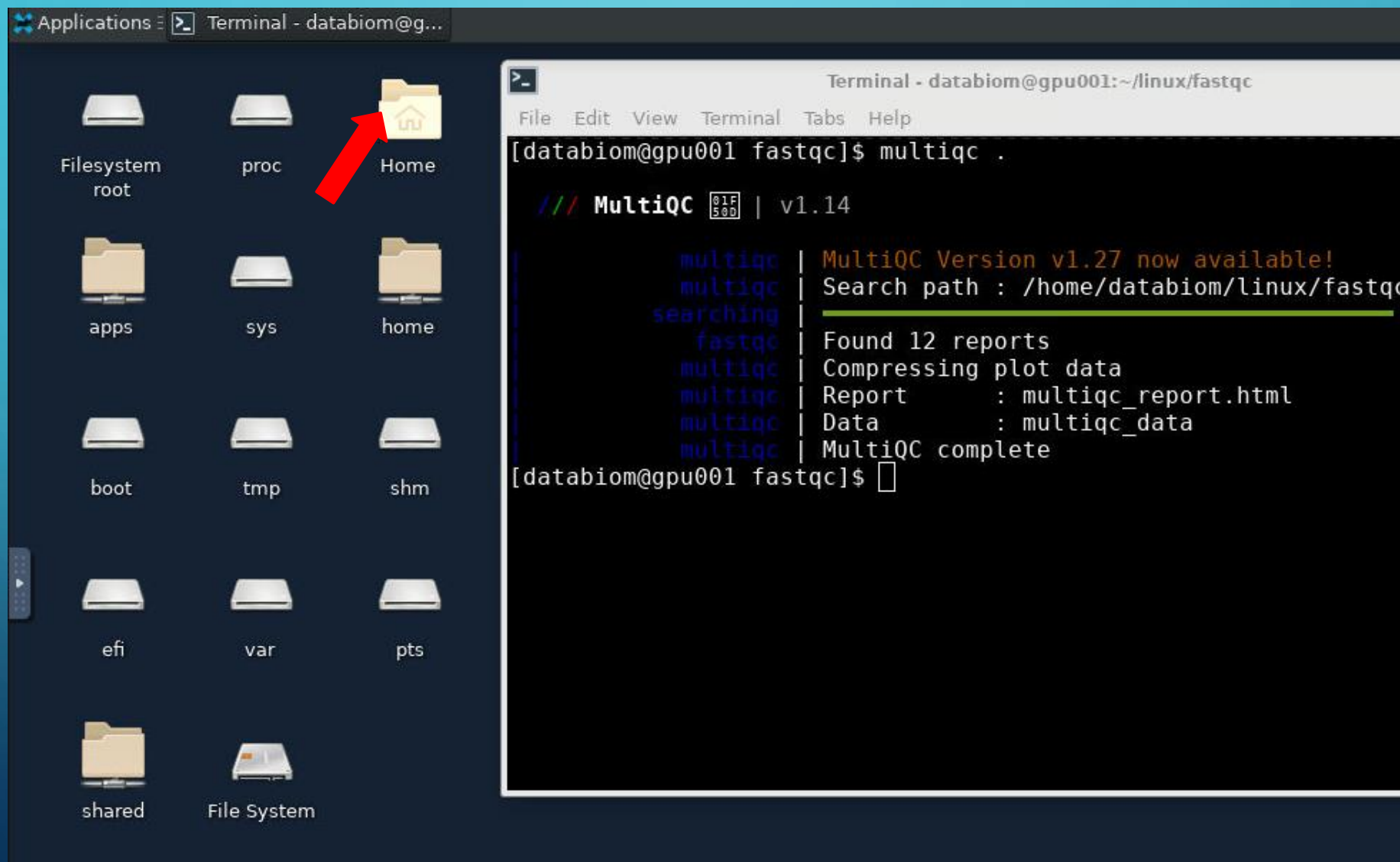
```
Terminal - databiom@gpu001:~/linux/fastqc
File Edit View Terminal Tabs Help
[databiom@gpu001 fastqc]$ multiqc .

/// MultiQC 01F500 | v1.14

multiqc | MultiQC Version v1.27 now available!
multiqc | Search path : /home/databiom/linux/fastqc
searching |  100% 24/24
fastqc | Found 12 reports
multiqc | Compressing plot data
multiqc | Report      : multiqc_report.html
multiqc | Data        : multiqc_data
multiqc | MultiQC complete
[databiom@gpu001 fastqc]$
```

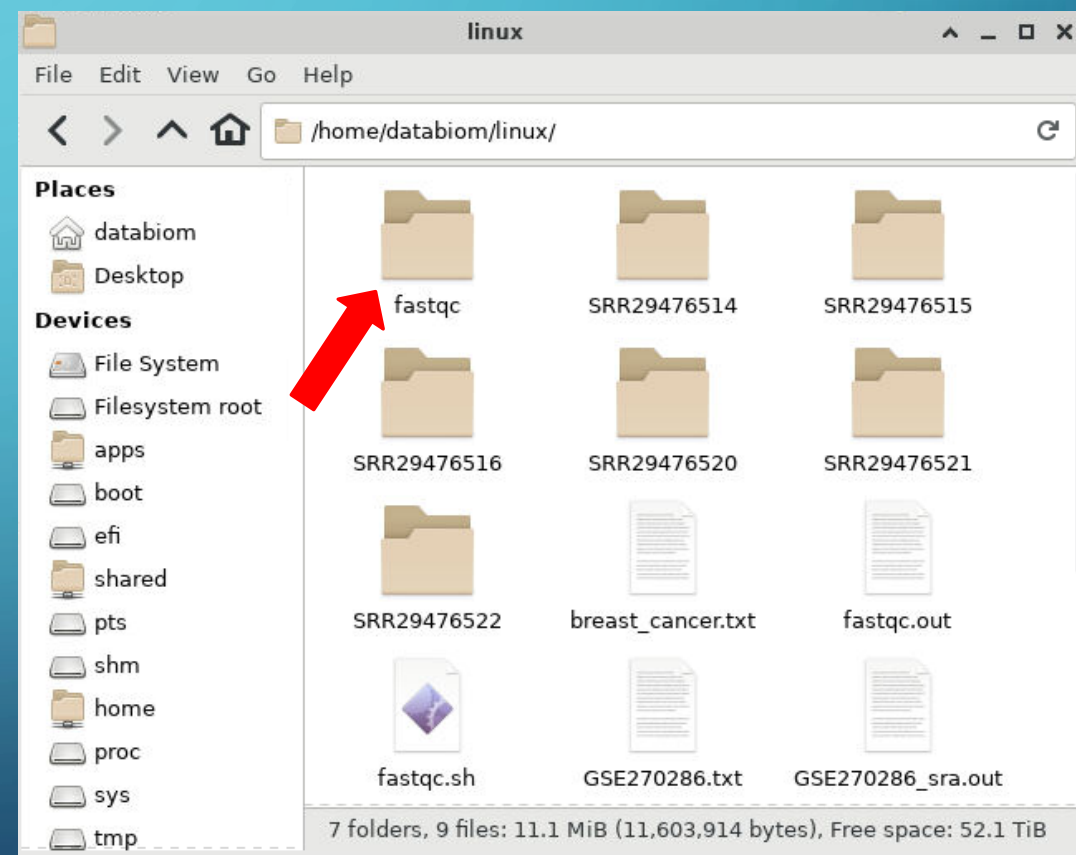
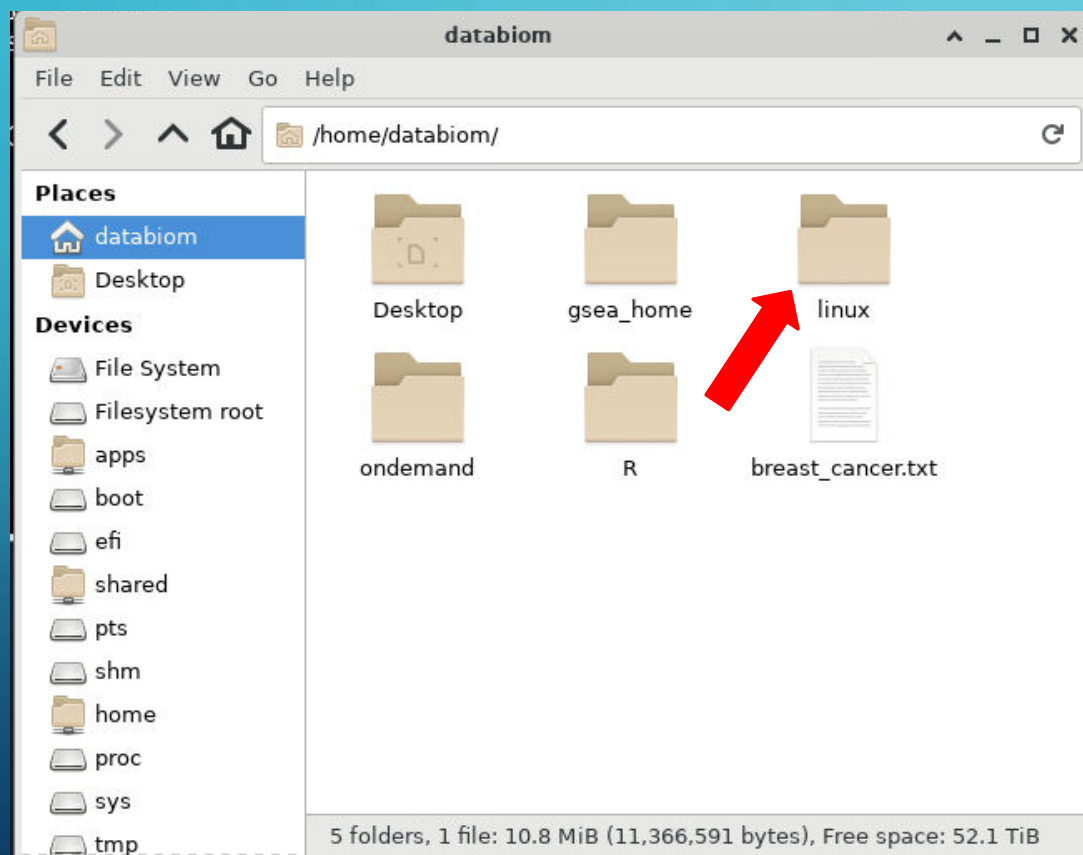
VIEW MULTIQC RESULTS

Click on 'Home' folder (you may need to move terminal window over →)



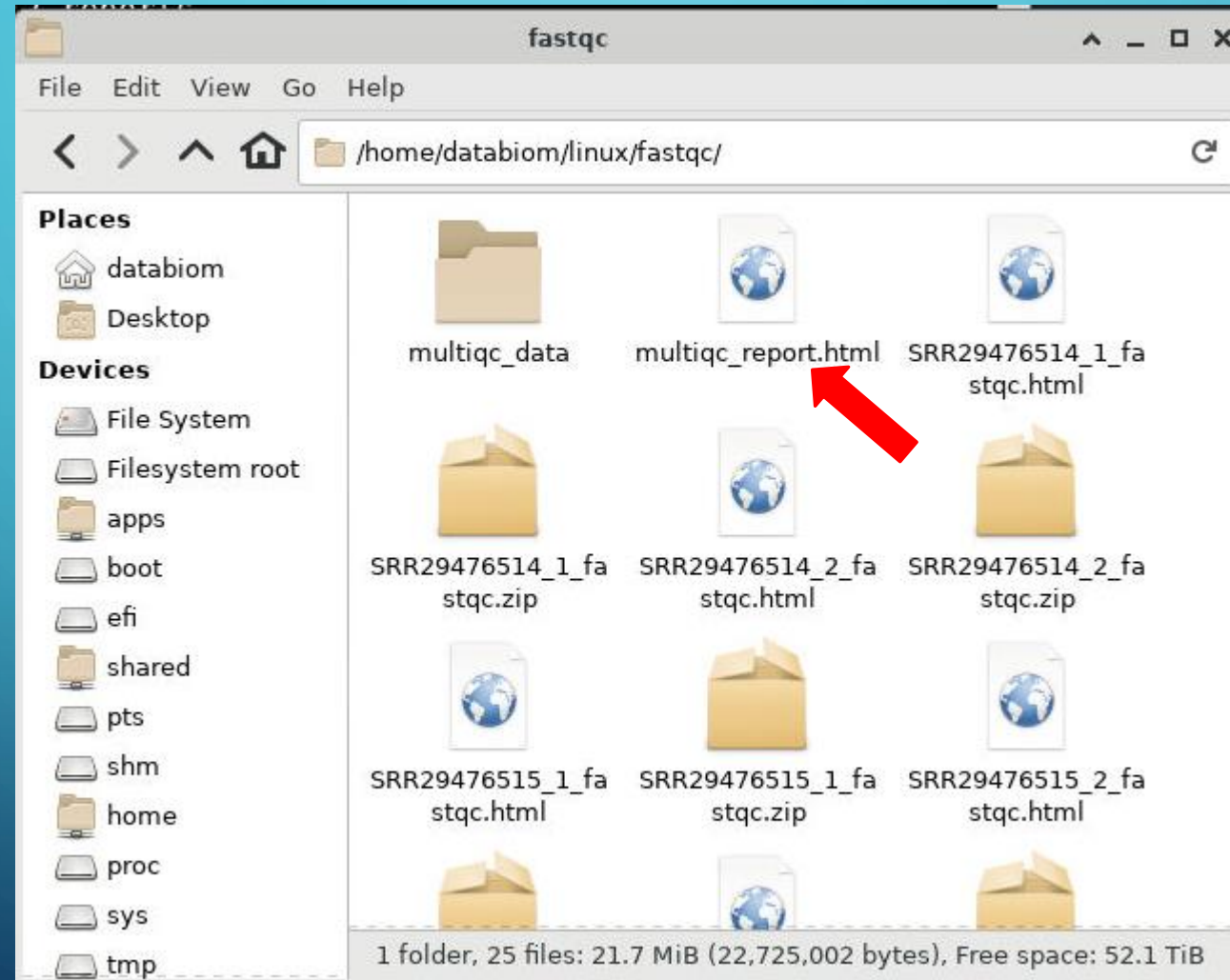
VIEW MULTIQC RESULTS

Click on the 'linux' folder and then the 'fastqc' folder:



VIEW MULTIQC RESULTS

Click on 'multiqc_report.html' file:



VIEW MULTIQC RESULTS

The file will open in a new browser window:

MultiQC v1.14

General Stats

FastQC

Sequence Counts

Sequence Quality Histograms

Per Sequence Quality Scores

Per Base Sequence Content

Per Sequence GC Content

Per Base N Content

Sequence Length Distribution

Sequence Duplication Levels

Overrepresented sequences

Adapter Content

Status Checks

MultiQC

A modular tool to aggregate results from bioinformatics analyses across many samples

Report generated on 2025-02-13, 15:24 CST based on data in: `/home/databiom/linux/fastqc`

Welcome! Not sure where to start? [Watch a tutorial video](#) (6:06)

General Statistics

[Copy table](#) [Configure Columns](#) [Plot](#) Showing 12/12 rows and 3/6 columns.

Sample Name	% Dups
SRR29476514_1	64.6%
SRR29476514_2	64.6%
SRR29476515_1	59.6%
SRR29476515_2	59.5%
SRR29476516_1	58.4%
SRR29476516_2	59.9%

GENERAL STATISTICS

General Statistics

 Copy table

 Configure Columns

 Plot

Showing 12¹/₁₂ rows and 3¹/₆ columns.

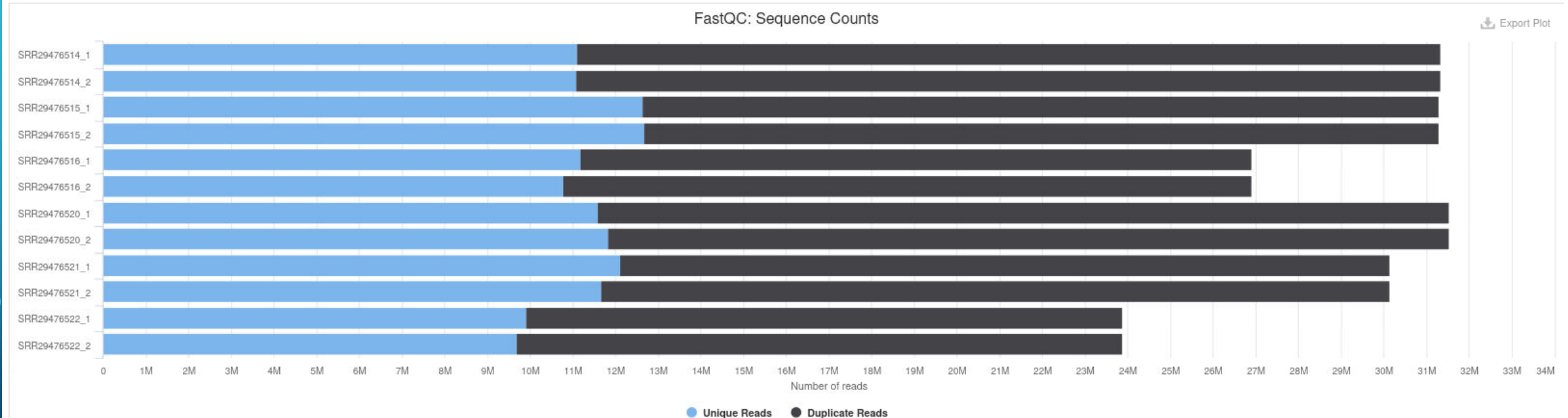
Sample Name	% Dups	% GC	M Seqs
SRR29476514_1	64.6%	51%	31.3
SRR29476514_2	64.6%	52%	31.3
SRR29476515_1	59.6%	50%	31.3
SRR29476515_2	59.5%	50%	31.3
SRR29476516_1	58.4%	50%	26.9
SRR29476516_2	59.9%	51%	26.9
SRR29476520_1	63.2%	50%	31.5
SRR29476520_2	62.5%	51%	31.5
SRR29476521_1	59.8%	50%	30.1
SRR29476521_2	61.2%	51%	30.1
SRR29476522_1	58.4%	51%	23.9
SRR29476522_2	59.4%	51%	23.9

NUMBER OF SEQUENCES (MILLIONS)

Sequence Counts

Sequence counts for each sample. Duplicate read counts are an estimate only.

Number of reads Percentages



In RNA-Seq libraries, sequences from different transcripts will be present at wildly different levels in the starting population. You will naturally tend to over-sequence parts of the library and therefore generate duplication and will therefore expect to see warnings or error from this module

QUALITY OF SEQUENCES

Sequence Quality Histograms

12

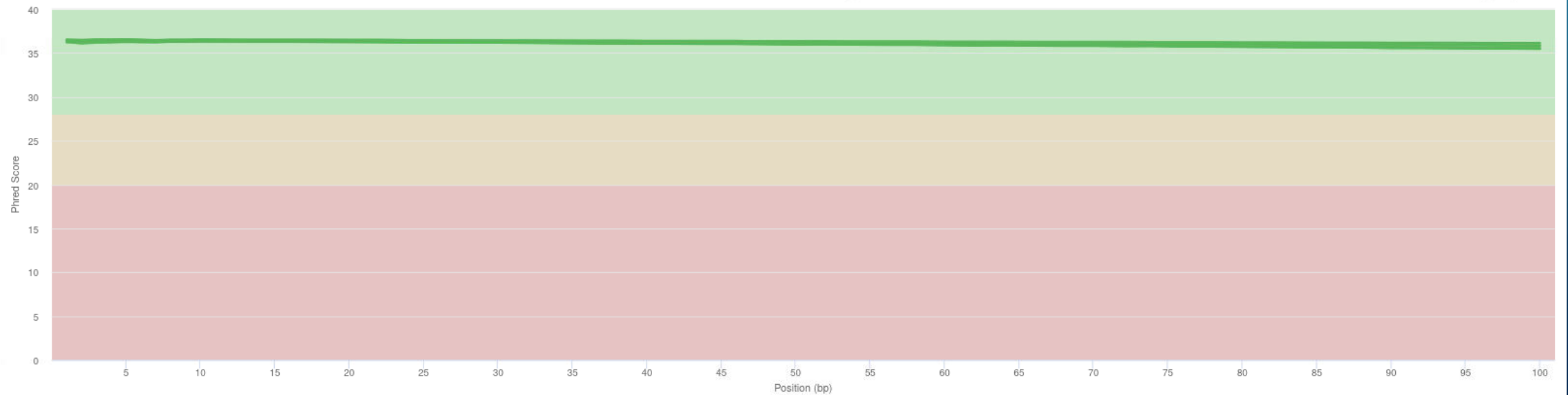
The mean quality value across each base position in the read.

Help

Y-Limits: ☒ on

FastQC: Mean Quality Scores

Export Plot



Created with MultiQC

Phred score > 30 is 99.90% accurate.

PER SEQUENCE QUALITY

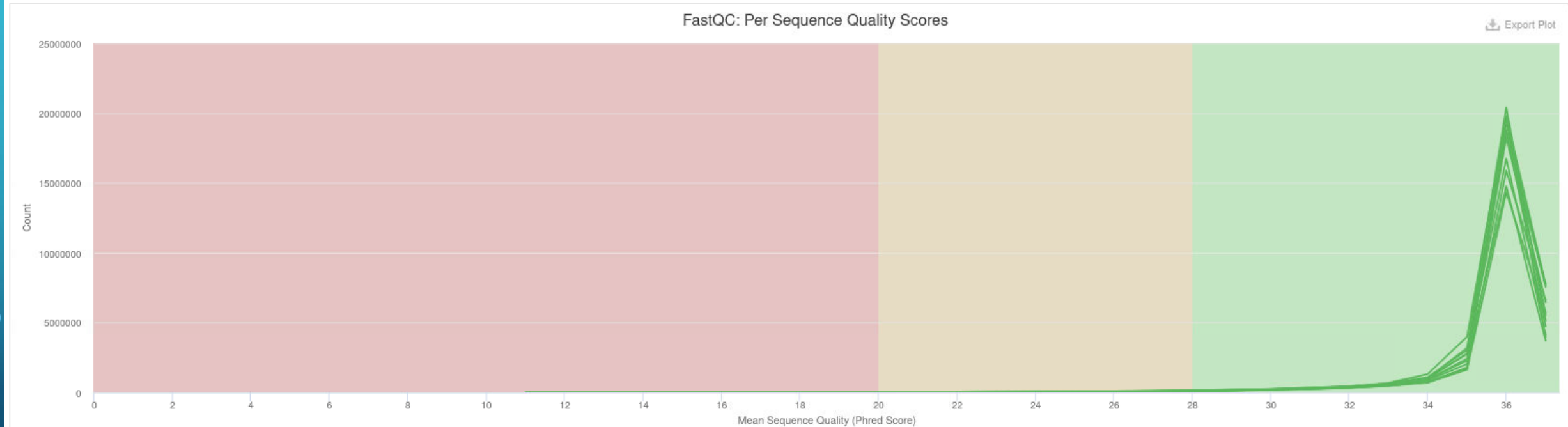
Per Sequence Quality Scores

12

The number of reads with average quality scores. Shows if a subset of reads has poor quality.

Help

Y-Limits: on



Phred score > 30 is 99.90% accurate.

PER BASE SEQUENCE CONTENT



It's worth noting that some types of library will always produce biased sequence composition, normally at the start of the read. Libraries produced by priming using random hexamers (including nearly all RNA-Seq libraries) inherit an intrinsic bias in the positions at which reads start.. Whilst this is a true technical bias, it isn't something which can be corrected by trimming and in most cases doesn't seem to adversely affect the downstream analysis. It will however produce a warning or error in this module.

GC CONTENT

Per Sequence GC Content

11

The average GC content of reads. Normal random library typically have a roughly normal distribution of GC content.

Help

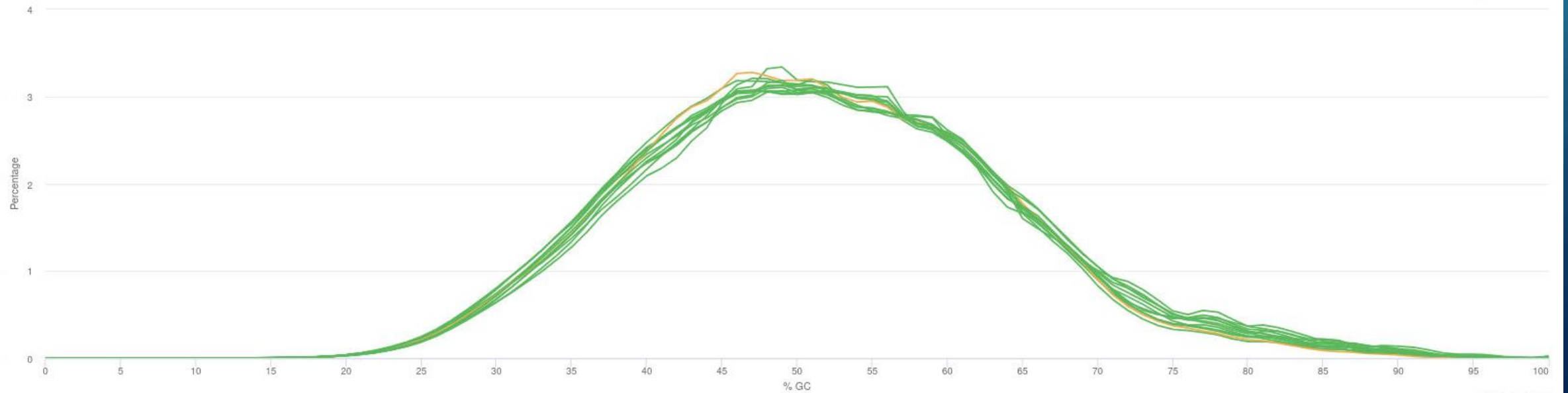
Y-Limits: ☒ on

Percentages

Counts

FastQC: Per Sequence GC Content

Export Plot



Created with MultiQC

PER BASE 'N' CALLS (UNKNOWN)

Per Base N Content

12

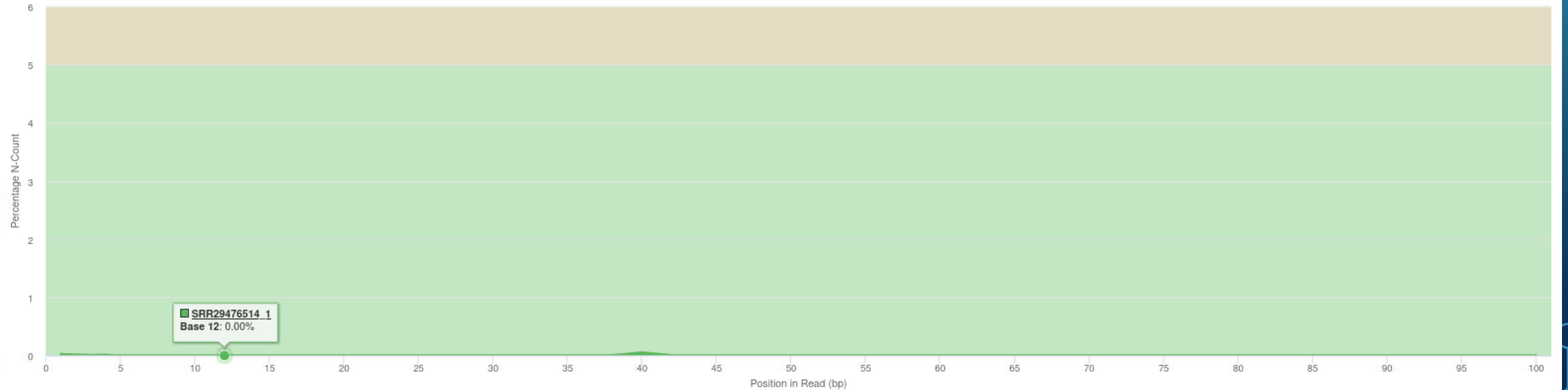
The percentage of base calls at each position for which an N was called.

Help

Y-Limits: on

FastQC: Per Base N Content

Export Plot



Sequence Length Distribution

12

All samples have sequences of a single length (101bp).

SEQUENCE DUPLICATION LEVELS

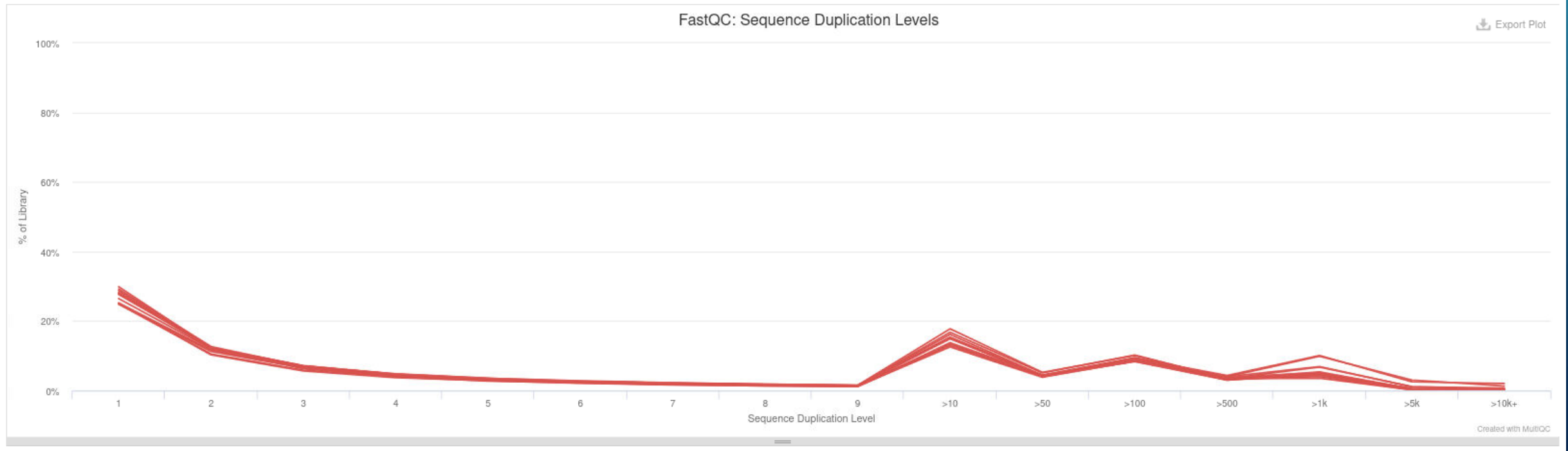
Sequence Duplication Levels

12

The relative level of duplication found for every sequence.

Help

Y-Limits: ☒ on



Overrepresented sequences

10

2

The total amount of overrepresented sequences found in each library.

12 samples had less than 1% of reads made up of overrepresented sequences

ADAPTER CONTENT OF SEQUENCES

Adapter Content

12

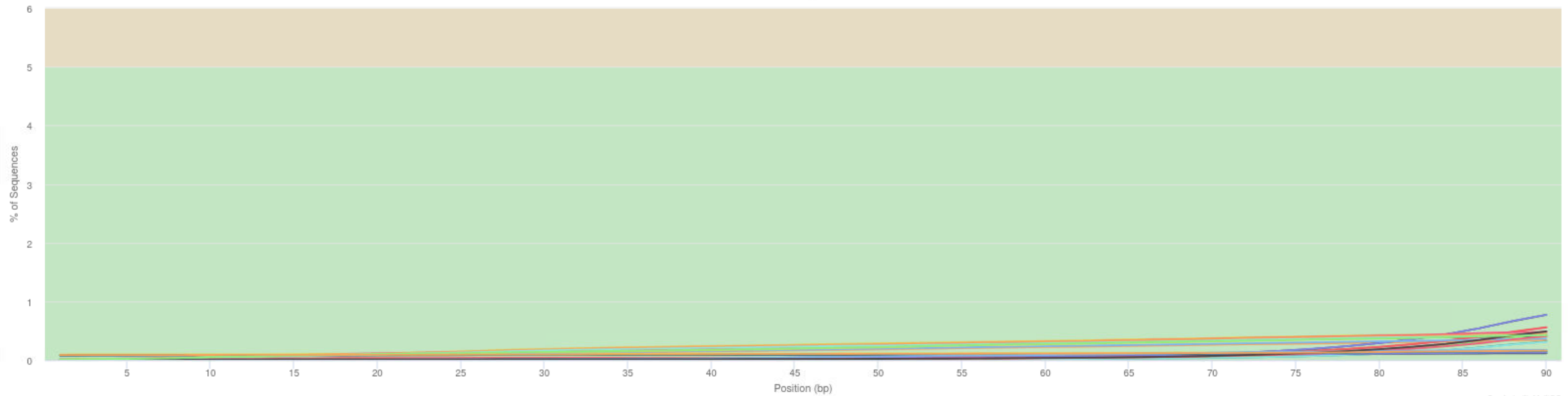
The cumulative percentage count of the proportion of your library which has seen each of the adapter sequences at each position.

Help

Y-Limits: on

FastQC: Adapter Content

Export Plot



Created with MultiQC

STATUS CHECKS

All things considered, these sequences are of good quality

Status Checks

Status for each FastQC section showing whether results seem entirely normal (green), slightly abnormal (orange) or very unusual (red).

Sort by highlight

Min: 0

Max: 1

FastQC: Status Checks

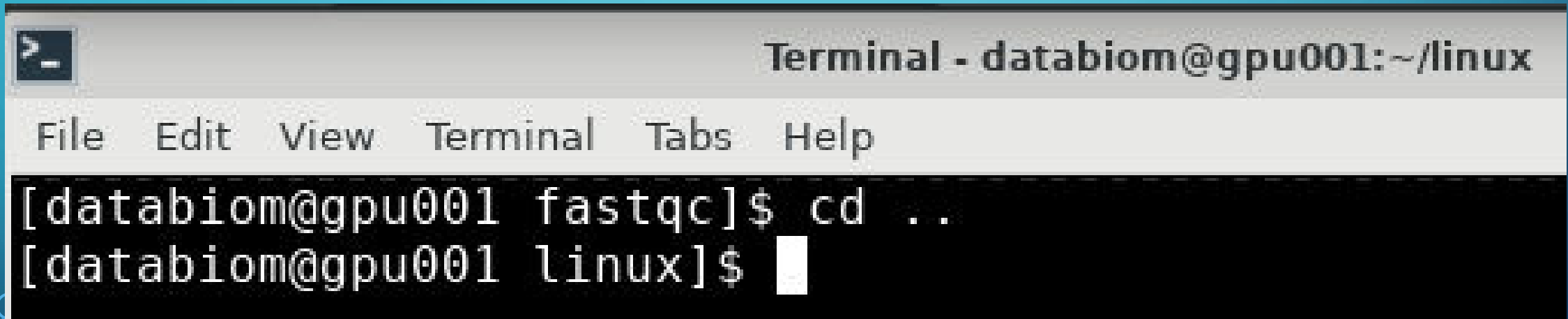
Export Plot



Created with MultiQC

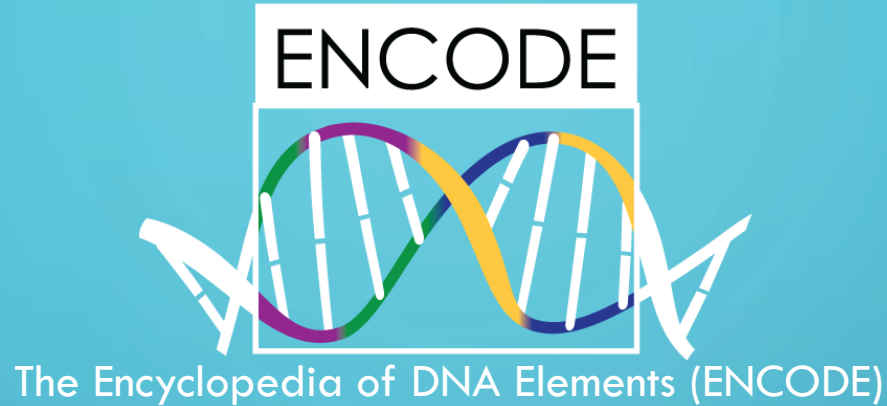
EXIT OUT OF THE MULTIQC BROWSER

Go back to the 'linux' directory

A terminal window titled "Terminal - databiom@gpu001:~/linux" with a menu bar containing "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal shows the command "cd .." being executed, changing the directory from "fastqc" to "linux".

```
>_ Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 fastqc]$ cd ..
[databiom@gpu001 linux]$
```

ALIGNMENT AND TRANSCRIPT QUANTIFICATION



ENCODE

Data

Encyclopedia

Materials & Methods

Help



Search...



Bulk RNA-seq Data Standards and Processing Pipeline

Regarding alignment and quantification:

The mapping of the reads is done using the STAR program (in some cases, both STAR and TopHat aligners are used to produce separate bam files) and the quantification of genes and transcripts is done with the RSEM program. Although there is general agreement between the mappings and the gene quantifications produced by different RNA-seq pipelines, quantifications of individual transcript isoforms, being much more complex, can differ substantially depending on the processing pipeline employed and are of unknown accuracy. Therefore, **alignments** and **gene quantifications** can be used confidently, while **transcript quantifications** should be used with care.

REFERENCE GENOME

Go to gencodegenes.org in a web browser



[Human](#)

[Mouse](#)

[How to access data](#)

[FAQ](#)

[Documentation](#)

[About us](#)

HUMAN

GENCODE 47 (October 2024)



MOUSE

GENCODE M36 (October 2024)



CHOOSE THE CORRECT GTF FILE

GTFs have the gene

GTF / GFF3 files

Content	Regions	Description	Download
Comprehensive gene annotation	CHR	It contains the comprehensive gene annotation on the reference chromosomes only	GTF GFF3
Comprehensive gene annotation	ALL	It contains the comprehensive gene annotation on the reference chromosomes, scaffolds, assembly patches and alternate loci (haplotypes)	GTF GFF3
Comprehensive gene annotation	PRI	It contains the comprehensive gene annotation on the primary assembly (chromosomes and scaffolds) sequence regions	GTF GFF3
Basic gene annotation	CHR	- It contains the basic gene annotation on the reference chromosomes only - This is a subset of the corresponding comprehensive annotation, including only those transcripts tagged as 'basic' in every gene - This is the main annotation file for most users	GTF GFF3
Basic gene annotation	ALL	- It contains the basic gene annotation on the reference chromosomes, scaffolds, assembly patches and alternate loci (haplotypes) - This is a subset of the corresponding comprehensive annotation, including only those transcripts tagged as 'basic' in every gene - This is a superset of the main annotation file	GTF GFF3
Basic gene annotation	<u>PRI</u>	- It contains the basic gene annotation on the primary assembly (chromosomes and scaffolds) sequence regions - This is a subset of the corresponding comprehensive annotation, including only those transcripts tagged as 'basic' in every gene	GTF GFF3

MAKE A NEW DIRECTORY FOR REFERENCE GENOME

Use the mkdir command to make a 'reference' directory and go into the newly created directory

```
[databiom@tsu-hpc01 linux]$ mkdir reference
```

```
[databiom@tsu-hpc01 linux]$ cd reference
```

DOWNLOAD GTF FILE

Use the wget command followed by the link you copied from the Gencode website (paste after the wget command and hit enter)

```
[databiom@tsu-hpc01 reference]$ wget https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_47/gencode.v47.primary_assembly.basic.annotation.gtf.gz
```

```
[databiom@tsu-hpc01 reference]$ wget https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_47/gencode.v47.primary_assembly.basic.annotation.gtf.gz
Will not apply HSTS. The HSTS database must be a regular and non-world-writable file.
ERROR: could not open HSTS store at '/home/databiom/.wget-hsts'. HSTS will be disabled.
--2025-02-14 12:56:52-- https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_47/gencode.v47.primary_assembly.basic.annotation.gtf.gz
Resolving ftp.ebi.ac.uk (ftp.ebi.ac.uk)... 193.62.193.165
Connecting to ftp.ebi.ac.uk (ftp.ebi.ac.uk)|193.62.193.165|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 33751722 (32M) [application/x-gzip]
Saving to: 'gencode.v47.primary_assembly.basic.annotation.gtf.gz'

gencode.v47.primary_assembly.basic.annotation 100%[=====>] 32.19M  2.15MB/s   in 18s

2025-02-14 12:57:10 (1.81 MB/s) - 'gencode.v47.primary_assembly.basic.annotation.gtf.gz' saved [33751722/33751722]

[databiom@tsu-hpc01 reference]$
```

You can also cp from the databiom account to your reference directory:

```
cp /home/databiom/linux/reference/gencode.v47.primary_assembly.basic.annotation.gtf.gz .
```

CHOOSE THE CORRECT FASTA FILE

FASTA files contain sequences of DNA

Fasta files

Content	Regions	Description	Download
Transcript sequences	CHR	Nucleotide sequences of all transcripts on the reference chromosomes	Fasta
Protein-coding transcript sequences	CHR	- Nucleotide sequences of coding transcripts on the reference chromosomes - Transcript biotypes: protein_coding, nonsense_mediated_decay, non_stop_decay, IG_*_gene, TR_*_gene, polymorphic_pseudogene, protein_coding_LoF	Fasta
Protein-coding transcript translation sequences	CHR	- Amino acid sequences of coding transcript translations on the reference chromosomes - Transcript biotypes: protein_coding, nonsense_mediated_decay, non_stop_decay, IG_*_gene, TR_*_gene, polymorphic_pseudogene, protein_coding_LoF	Fasta
Long non-coding RNA transcript sequences	CHR	Nucleotide sequences of long non-coding RNA transcripts on the reference chromosomes	Fasta
Genome sequence (GRCh38.p14)	ALL	- Nucleotide sequence of the GRCh38.p14 genome assembly version on the reference chromosomes, scaffolds, assembly patches and haplotypes - The sequence region names are the same as in the GTF/GFF3 files	Fasta
Genome sequence, primary assembly (GRCh38)	<u>PRI</u>	- Nucleotide sequence of the GRCh38 primary genome assembly (chromosomes, scaffolds, assembly patches and haplotypes) - The sequence region names are the same as in the GTF/GFF3 files	Fasta

DOWNLOAD FASTA FILE

Use the `wget` command followed by the link you copied from the Gencode website for the FASTA file (paste after the `wget` command and hit enter)

```
[databiom@tsu-hpc01 reference]$ wget https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_47/GRCh38.primary_assembly.genome.fa.gz

[databiom@gpu001 reference]$ wget https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_47/GRCh38.primary_assembly.genome.fa.gz
Will not apply HSTS. The HSTS database must be a regular and non-world-writable file.
ERROR: could not open HSTS store at '/home/databiom/.wget-hsts'. HSTS will be disabled.
--2025-02-16 22:19:37-- https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_47/GRCh38.primary_assembly.genome.fa.gz
Resolving ftp.ebi.ac.uk (ftp.ebi.ac.uk)... 193.62.193.165
Connecting to ftp.ebi.ac.uk (ftp.ebi.ac.uk)|193.62.193.165|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 845635028 (806M) [application/x-gzip]
Saving to: 'GRCh38.primary_assembly.genome.fa.gz'

GRCh38.primary_assembly.genome.fa.gz  100%[=====>] 806.46M  2.08MB/s  in 5m 25s

2025-02-16 22:25:04 (2.48 MB/s) - 'GRCh38.primary_assembly.genome.fa.gz' saved [845635028/845635028]

[databiom@gpu001 reference]$
```

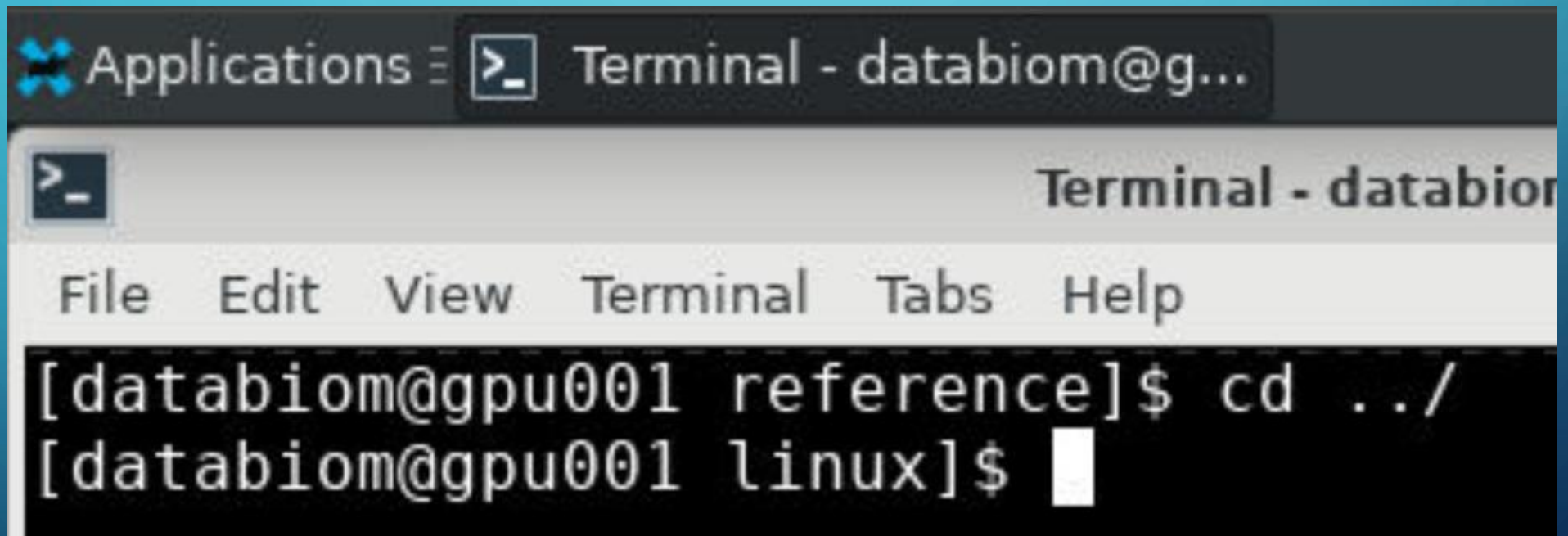


You can also `cp` from the databiom account to your reference directory:

```
cp /home/databiom/linux/reference/GRCh38.primary_assembly.genome.fa.gz .
```

COPY OVER A SCRIPT FOR THE ALIGNMENT

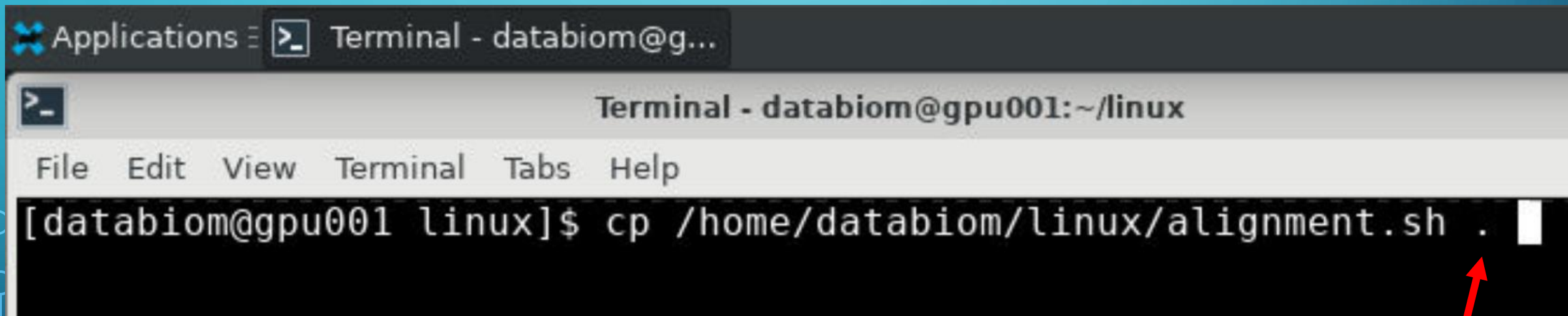
First, go back to the 'linux' directory using the cd command

A screenshot of a terminal window titled 'Terminal - databiom@g...'. The window has a menu bar with 'File', 'Edit', 'View', 'Terminal', 'Tabs', and 'Help'. The terminal content shows the user at the prompt '[databiom@gpu001 reference]\$' typing 'cd ../'. The next line shows the prompt has changed to '[databiom@gpu001 linux]\$' with a cursor. The window title bar also shows 'Terminal - databior' on the right side.

```
Applications ▢ >_ Terminal - databiom@g...  
>_ Terminal - databior  
File Edit View Terminal Tabs Help  
[databiom@gpu001 reference]$ cd ../  
[databiom@gpu001 linux]$
```

COPY OVER A SCRIPT

Copy the alignment.sh script to your 'linux' directory

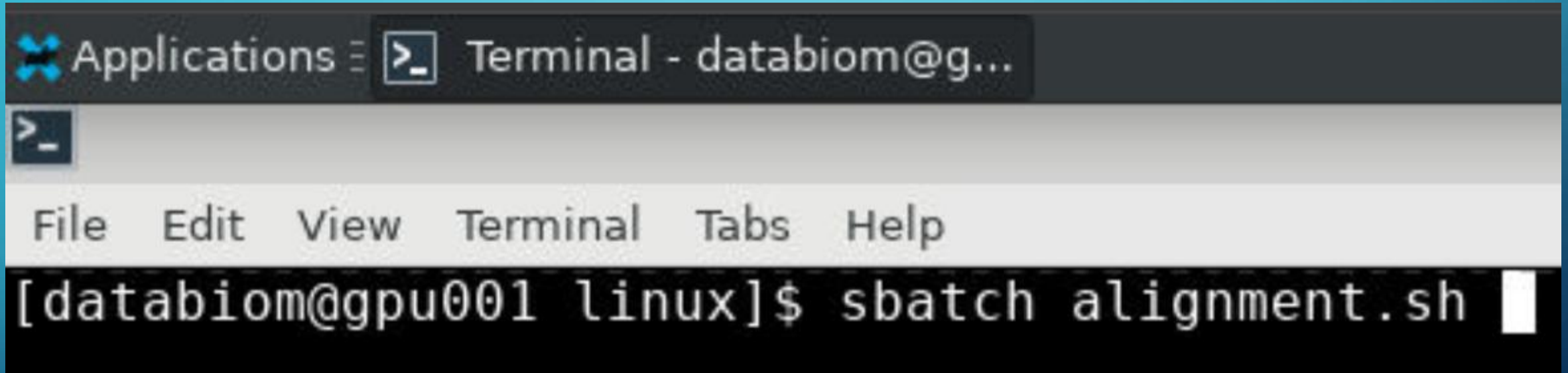


A screenshot of a terminal window. The title bar shows 'Applications' and 'Terminal - databiom@g...'. The terminal header displays 'Terminal - databiom@gpu001:~/linux'. Below the header is a menu bar with 'File', 'Edit', 'View', 'Terminal', 'Tabs', and 'Help'. The main terminal area shows a command prompt '[databiom@gpu001 linux]\$' followed by the command 'cp /home/databiom/linux/alignment.sh .' and a white cursor at the end. A red arrow points from the text 'Don't forget' to the period at the end of the command.

```
Applications ▢ Terminal - databiom@g...  
Terminal - databiom@gpu001:~/linux  
File Edit View Terminal Tabs Help  
[databiom@gpu001 linux]$ cp /home/databiom/linux/alignment.sh .
```

Don't forget

SUBMIT THE ALIGNMENT.SH SCRIPT

A screenshot of a terminal window titled "Terminal - databiom@g...". The window has a menu bar with "File", "Edit", "View", "Terminal", "Tabs", and "Help". The prompt is "[databiom@gpu001 linux]\$". The command "sbatch alignment.sh" has been entered, and a white cursor is at the end of the line.

```
Applications ▢ >_ Terminal - databiom@g...  
>_  
File Edit View Terminal Tabs Help  
[databiom@gpu001 linux]$ sbatch alignment.sh
```

We'll go over the contents of the script next time...



GOOD JOB!

RNA-SEQ: PART 1